EDGE-BASED REAL-TIME PATIENT MONITORING SYSTEM

Presented by :

SOMEN PRADHAN

Regd No.: 2301298542

Roll No.: 59

Subject Faculty: Jyoti Ranjan Swain

ABSTRACT

- Edge-Based Real-Time Patient Monitoring System is an AI-based healthcare solution.
- Real-time patient monitoring using AI and Edge Computing to predict critical or stable condition.
- Monitors vital signs such as Heart Rate, SpO_2, Temperature, Respiration Rate, and Blood Pressure.
- Uses machine learning for patient condition prediction.
- Multiple medical datasets are used for analysis and monitoring.
- Provides fast decision support for doctors.
- Improves patient safety and healthcare efficiency.

INTRODUCTION

- AI-based ICU monitoring system for healthcare efficiency and patient safety
- Continuous tracking of vital signs (Heart Rate, SpO_2, Temperature, Respiration, Blood Pressure)
- Reduces delays in emergency situations
- Uses Machine Learning for patient condition prediction
- Shows Stable or Critical status
- Helps doctors make faster decisions and improves patient safety

PROBLEM STATEMENT

- ICU patients require continuous monitoring of vital signs.
- Manual monitoring systems are slow and error-prone.
- Manual monitoring is slow; AI-based monitoring provides faster and accurate results.
- Delays in detecting critical conditions can risk patient life.
- Doctors need a faster and automated monitoring system.
- Need for a smart system to identify Stable or Critical patients quickly.

TECH STACK

Frontend

- HTML, CSS, JavaScript
- React

Backend

- FastAPI
- Python

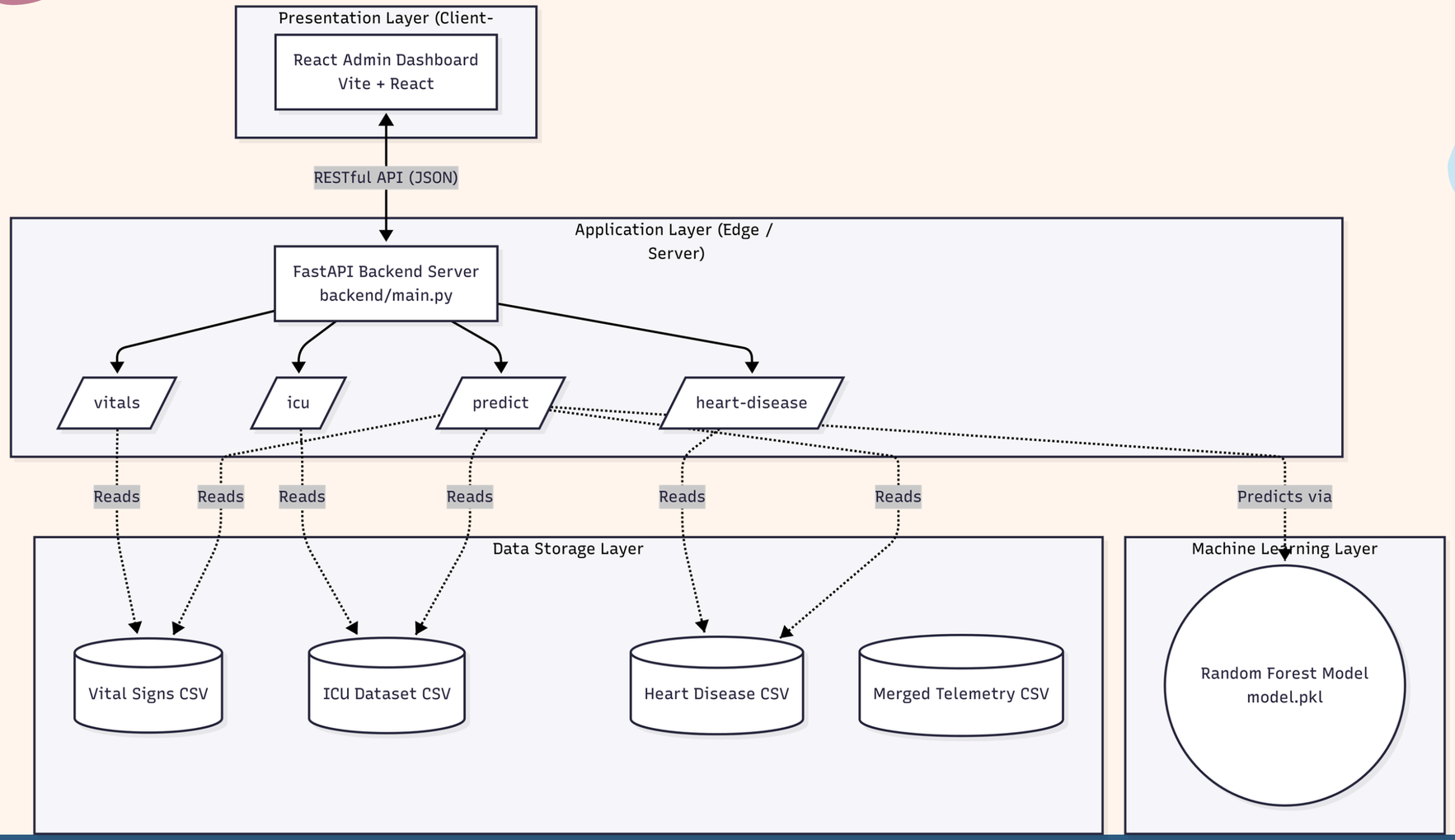
Machine Learning

- Scikit-learn
- Random Forest
- Pickle Model

Data

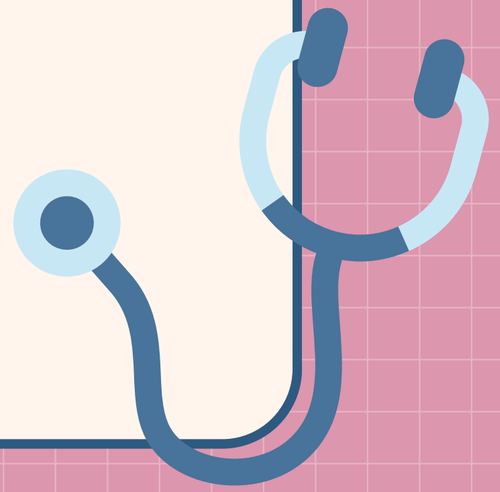
- CSV Datasets (heart disease, icu, vital signs)

TECHNICAL FLOW



BENEFITS

- Real-time patient monitoring
- Fast and accurate prediction
- Improved healthcare efficiency
- Early detection of critical conditions
- Reduced manual monitoring effort
- Better decision making for doctors
- Easy and user-friendly ICU dashboard
- Low-cost and scalable system



FUTURE SCOPE

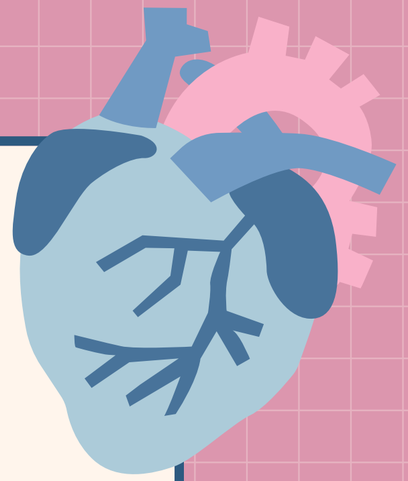
- IoT sensor integration for real-time data collection
- Cloud-based patient monitoring system
- Mobile application for doctors and hospitals
- Real-time ICU device connectivity
- Advanced AI models for better prediction
- Integration with hospital databases
- Remote patient monitoring system

CONCLUSION

- AI-based ICU monitoring improves patient safety and decision-making
- Real-time monitoring helps detect critical conditions early
- Machine learning enables fast and accurate prediction
- Reduces manual monitoring and human errors
- Improves healthcare efficiency and patient care

REFERENCES

- PhysioNet Vital Signs Dataset
- ICU Patient Dataset (Public Medical Dataset)
- Heart Disease Dataset (UCI / Kaggle)
- Scikit-learn Documentation
- FastAPI Documentation
- Python Official Documentation
- Machine Learning Research Papers on ICU Monitoring



THANK YOU

Do you have Any Dought?